Real Numbers: Mini-Lessons

NC.8.NS.1–2 · NC.8.EE.2

Name _____ Date _____ Period _____

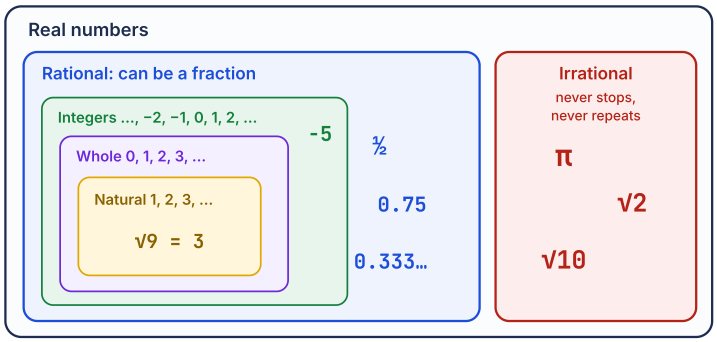
1.1 Rational or irrational? NC.8.NS.1

A number is **rational** if it can be written as a **fraction** of two integers. Its decimal **stops** or **repeats**. A number is **irrational** if its decimal **never stops and never repeats**, like π or √2.

Terminating decimal
a decimal that stops: 0.75

Repeating decimal
a decimal with digits that repeat forever: 0.333...

Perfect square
a whole number times itself: 1, 4, 9, 16, 25, 36, 49, ...



Square roots: √(perfect square) is rational (√49 = 7). √(not a perfect square) is irrational (√10).
Say it this way: "___ is rational because it can be written as a fraction."

Watch out: A long decimal is not always irrational. If it **repeats**, it is rational.

Walk through: Is √49 rational or irrational?

1. Is 49 a perfect square?

2. So what is √49?

3. Can 7 be written as a fraction?

Check your understanding

1 Score 3 NC.8.NS.1

Is 0.333… rational or irrational?

Ⓐ Rational Ⓑ Irrational

2 Score 3 NC.8.NS.1

Is √10 rational or irrational?

Ⓐ Irrational Ⓑ Rational

3 Score 3 NC.8.NS.1

Which number is **irrational**?

(A) π (B) −5

(C) 0.75 (D) √16

1.2 Repeating decimals to fractions (the 9's rule) NC.8.NS.1

To change a repeating decimal into a fraction: the **repeating digits go on top**. The **bottom is all 9s**, one 9 for each repeating digit. Then **simplify**.

Bar notation

a bar over the digits that repeat: $0.\overline{3} = 0.333...$

Simplify

divide the top and bottom by the same number

$0.272727\ldots \qquad 0.\overline{27}$

$$\frac{27}{99} \div 9 = \frac{3}{11}$$

2 repeating digits → two 9s

Decimal	Repeating digits	Fraction	Simplified
$0.\overline{3}$	1 digit	3/9	1/3
$0.\overline{27}$	2 digits	27/99	3/11
$0.\overline{125}$	3 digits	125/999	125/999

Watch out: Count the 9s carefully: one 9 for **each** repeating digit. And always simplify.

Walk through: Write 0.363636… as a fraction

1 Which digits repeat?

2 What is the bottom of the fraction?

3 Simplify 36/99. What number divides both?

Check your understanding

1 Score 3 NC.8.NS.1

Write $0.\overline{5}$ as a fraction.

(A) 5/10 (B) 5/9

(C) 1/5 (D) 5/99

2 Score 3 NC.8.NS.1

Write $0.\overline{12}$ as a fraction in simplest form.

(A) 12/9 (B) 4/33

(C) 3/25 (D) 12/99

1.3 Square roots and cube roots NC.8.EE.2

A **square root** undoes squaring: $\sqrt{25} = 5$ because $5 \times 5 = 25$. A **cube root** undoes cubing: $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$.

Squared (^{2})

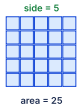
a number times itself: $6^2 = 6 \times 6 = 36$

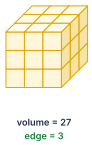
Cubed (^{3})

a number times itself 3 times: $2^3 = 2 \times 2 \times 2 = 8$

$\sqrt{}$ and $\sqrt[3]{}$

square root and cube root: they go backward





n	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	125	216	343	512	729	1000

Watch out: $\sqrt{}$ does **not** mean "divide by 2", and $\sqrt[3]{}$ does **not** mean "divide by 3". $\sqrt{16} = 4$, not 8.

Walk through: Find $\sqrt[3]{125}$

1 What does $\sqrt[3]{125}$ ask?

2 Try 5. What is $5 \times 5 \times 5$?

3 So $\sqrt[3]{125} = ?$

Check your understanding

1 Score 3 NC.8.EE.2

Find $\sqrt{81}$.

Answer: _____

2 Score 3 NC.8.EE.2

Find $\sqrt[3]{8}$.

Answer: _____

3 Score 3 NC.8.NS.1

Which number is a **perfect square**?

(A) 50 (B) 18

(C) 27 (D) 36

1.4 Estimate roots and order numbers NC.8.NS.2

To estimate a square root that is not a perfect square, find the **perfect squares just below and just above**. The root is **between** their roots. To compare numbers, write them all as **decimals** first.

Estimate

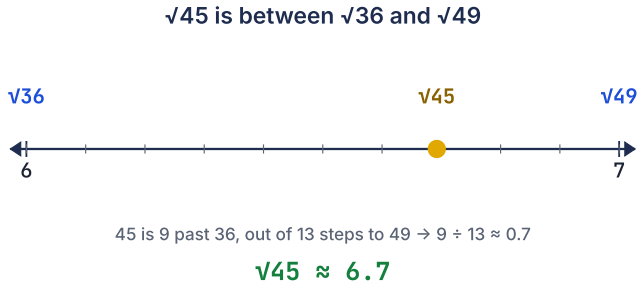
a close guess you can explain

Between

√45 is between 6 and 7 because 36 < 45 < 49

Least to greatest

smallest first, biggest last



4-step estimate of √45: 36 < 45 < 49 → (45 − 36) ÷ (49 − 36) = 9 ÷ 13 ≈ 0.7 → √45 ≈ 6.7

Watch out: Line up the decimals before you compare. 3.5 is bigger than 3.375 even though "375" looks long.

Walk through: `Estimate √20 to the nearest tenth`

1. Which perfect squares are just below and just above 20?

_____

2. So √20 is between...

_____

3. Find the tenths: (20 − 16) ÷ (25 − 16) = ?

_____

Check your understanding

1 Score 3 NC.8.NS.2

$\sqrt{30}$ is between which two whole numbers?

(A) 5 and 6
(B) 15 and 16
(C) 6 and 7
(D) 25 and 36

2 Score 3 NC.8.NS.2

Which is greater: $\sqrt{50}$ or 7.5?

(A) 7.5
(B) √50
(C) They are equal

3 Score 3 NC.8.NS.2

Order from least to greatest: $\sqrt{5}, 2.1, 7/3$

(A) √5, 2.1, 7/3
(B) 2.1, 7/3, √5
(C) 7/3, √5, 2.1
(D) 2.1, √5, 7/3